中国人民大学
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Mountain View
A Blossom of Conditional Diffusion Models

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2022-11-02

Content

- Background
- Selected Papers
- Industrial Application
- Tools for Research
- Issues

Background

- The start of modern use: DDPM[1]

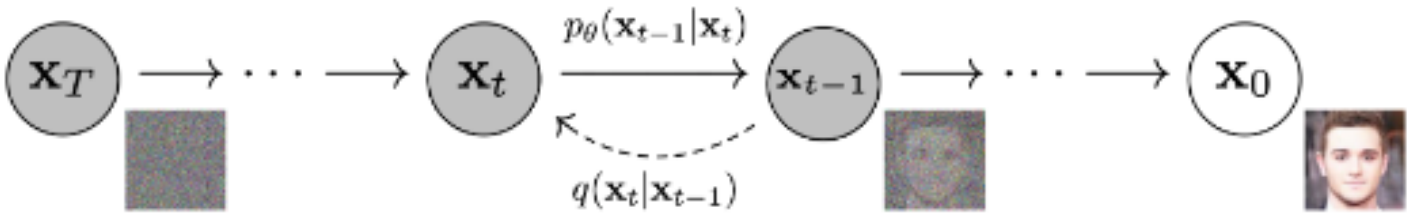


Figure from DDPM paper (https://arxiv.org/abs/2006.11239).

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \qquad q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

$$p_\theta(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t), \qquad p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

Background

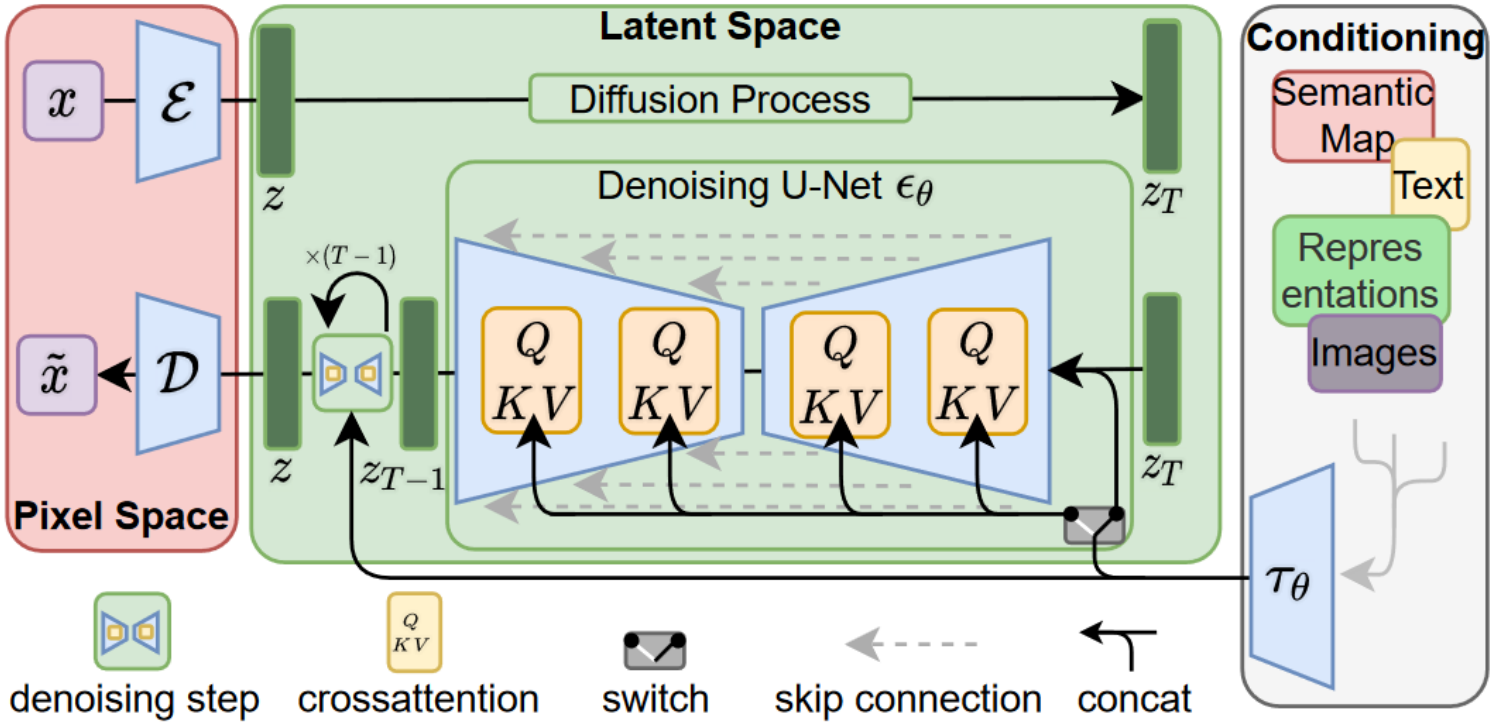
- DMs catch on.
- 100+ submissions in ICLR 2023(total 4966).
- DMs made excellent business success.

Keywords	2022	2023	rank ↑
large language model	208	32	176
diffusion model	173	14	159
offline reinforcement learning	59	20	39
sparsity	85	49	36
adversarial training	19	47	-28
differential privacy	45	23	22
fairness	43	22	21
model compression	61	41	20
domain generalization	55	36	19
time series	58	40	18

Data from https://zhuanlan.zhihu.com/p/570371424

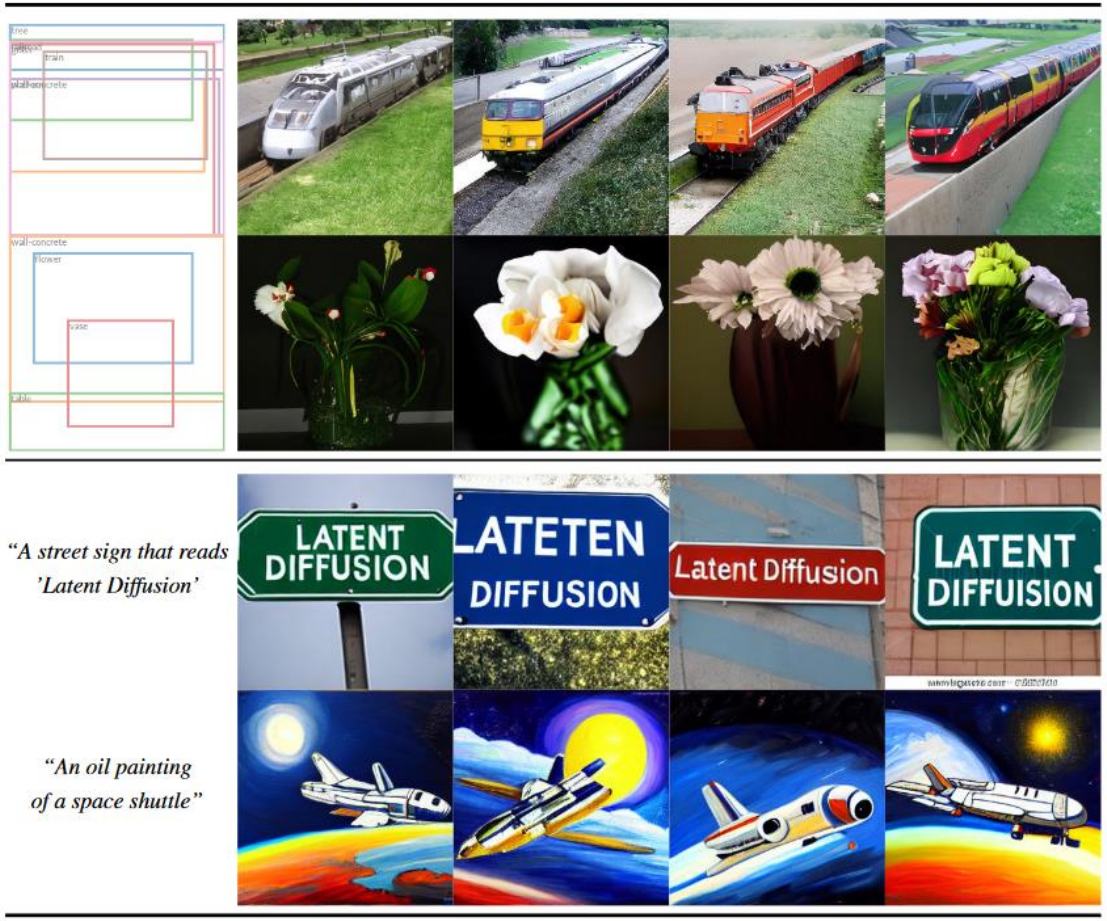
Selected Papers

- A representative root of Conditional Diffusion Models: Latent-Diffusion[2]



Selected Papers

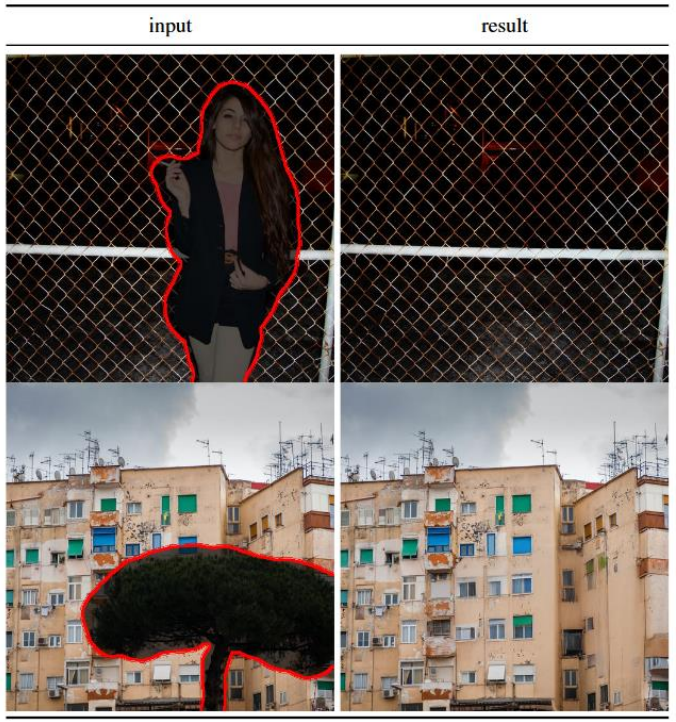
- Latent-Diffusion is trained on 4000+ A100 GPUs for:
- Unconditional Image Generation
- Text to Image
- Super-Resolution
- Inpainting
- Semantic Synthesis



Selected Papers

Super-Resolution

Semantic Synthesis



Inpainting

Selected Papers

Attempts of CDMs on various tasks:

Txt-2-Img:

- Imagen[3](Google)
- DALLE-2[4], GLIDE[5](Open-ai)
- VQ-Diffusion[6], Frido[7](Microsoft)

Txt-2-Speech:

- ProDif[8](ZJU)

Txt-2-Motion:

- MDM[9](Tel Aviv University)

Txt-2-Art:

- DYAD[10](CAS)

This is a colorful painting of a landscape in the style of cubism drawn by David Burliuk.

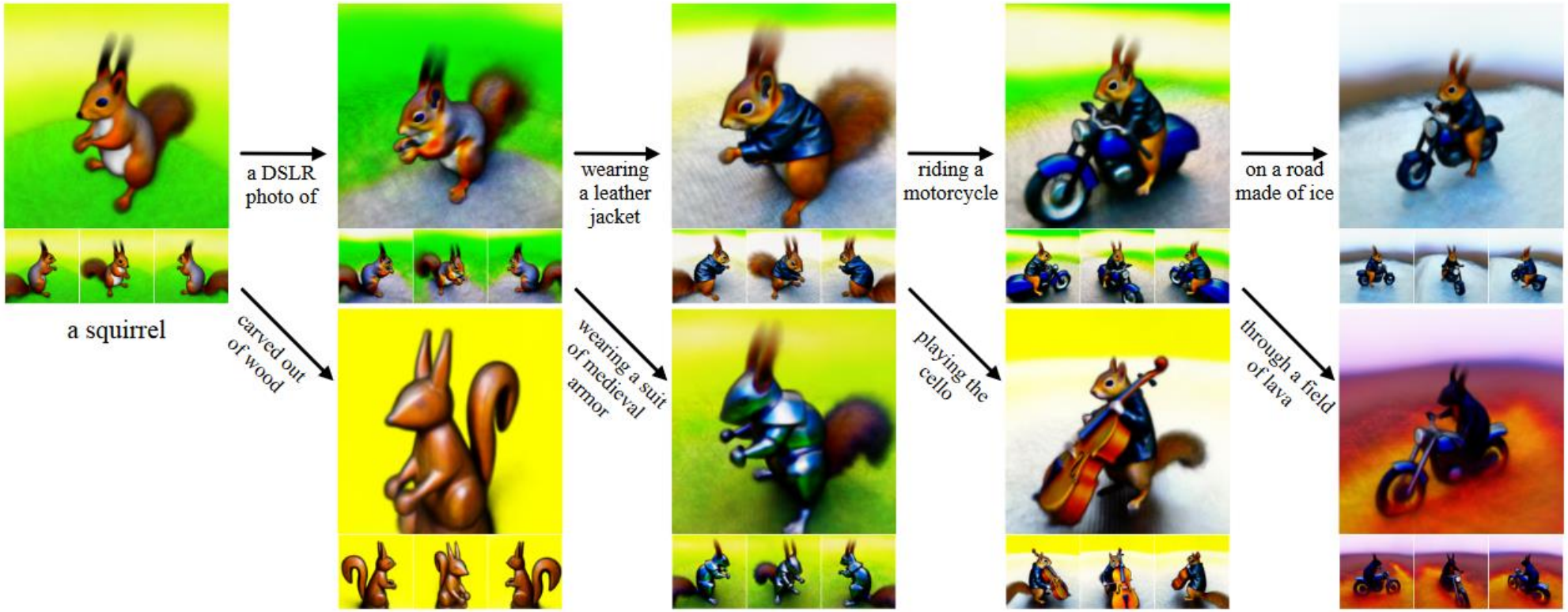
Picasso's painting of a cityscape in cubism style.

A romantic dream-like oil painting by Ivan Aivazovsky of cloudscape.

Selected Papers

Attempts of CDMs on various tasks:

Txt-2-3D: DreamFusion[11] (Google)



Selected Papers

Attempts of CDMs on various tasks:

Txt-2-Video: Make-A-Video[12] (Meta)

(a) A dog wearing a superhero outfit with red cape flying through the sky.

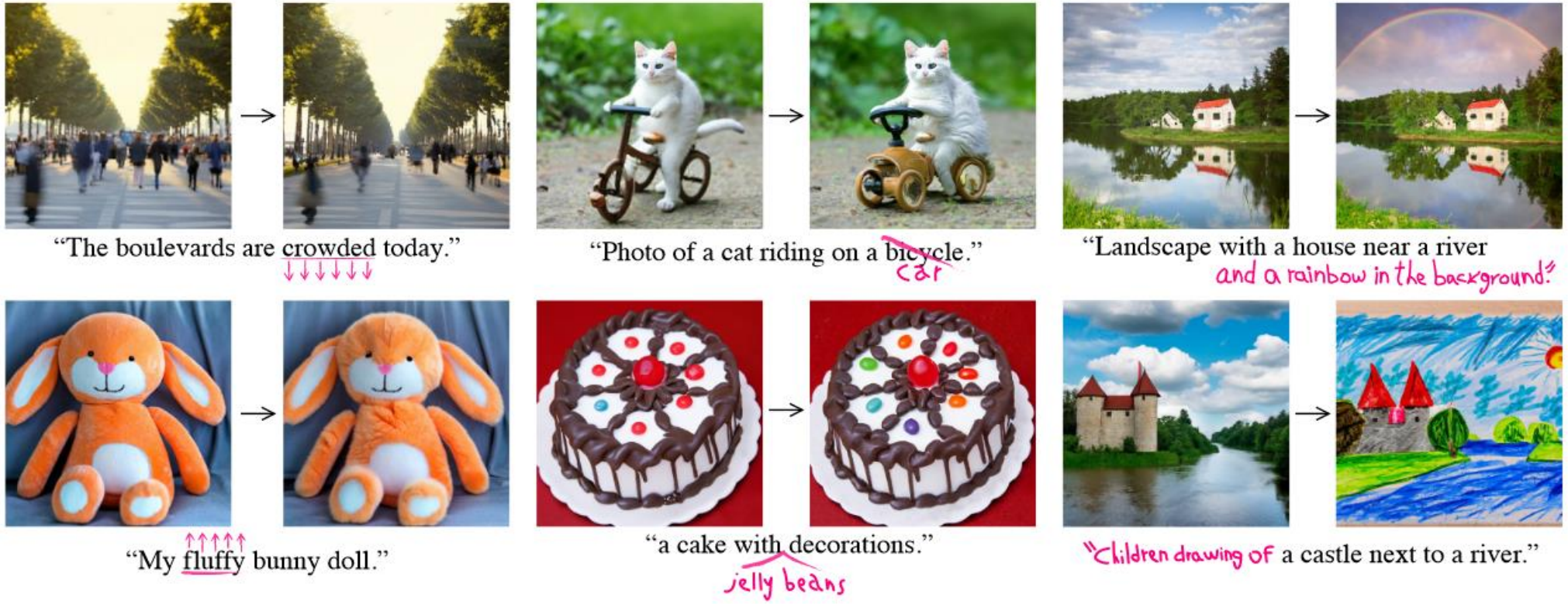
(b) There is a table by a window with sunlight streaming through illuminating a pile of books.

(c) Robot dancing in times square.

Selected Papers

Attempts of CDMs on Image Editing: How people flood in a hot direction

- Prompt-to-Prompt[13](Google): Editing only prompts.

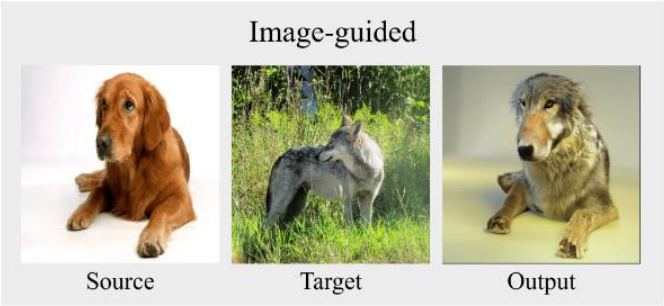


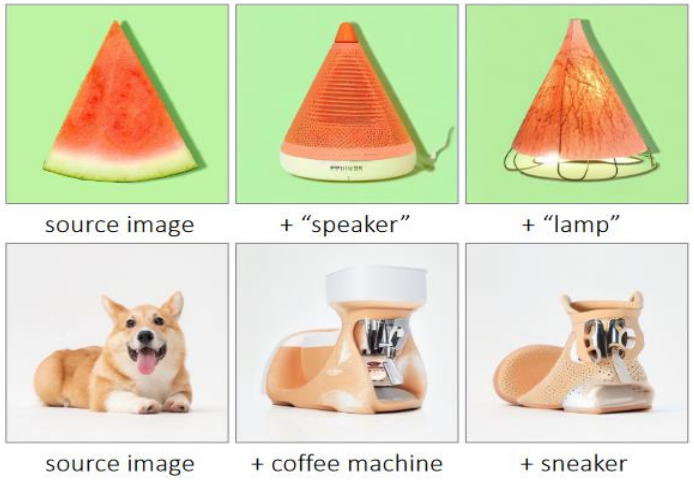
Selected Papers

Attempts of CDMs on Image Editing:
How people flood in a hot direction

- Text and Image guided[14](KAIST)
  Work for image guidance as well.

- MagicMix[15](TikTok)



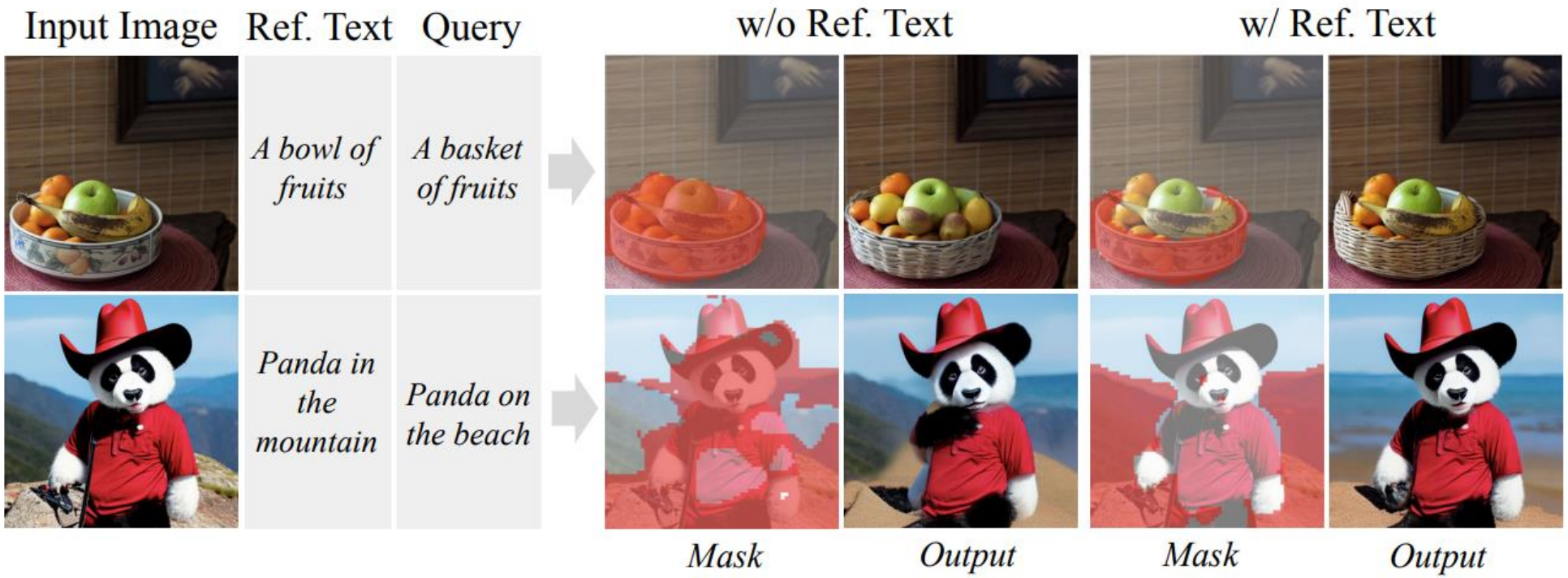


Novel objects.

Selected Papers

Attempts of CDMs on Image Editing: How people flood in a hot direction

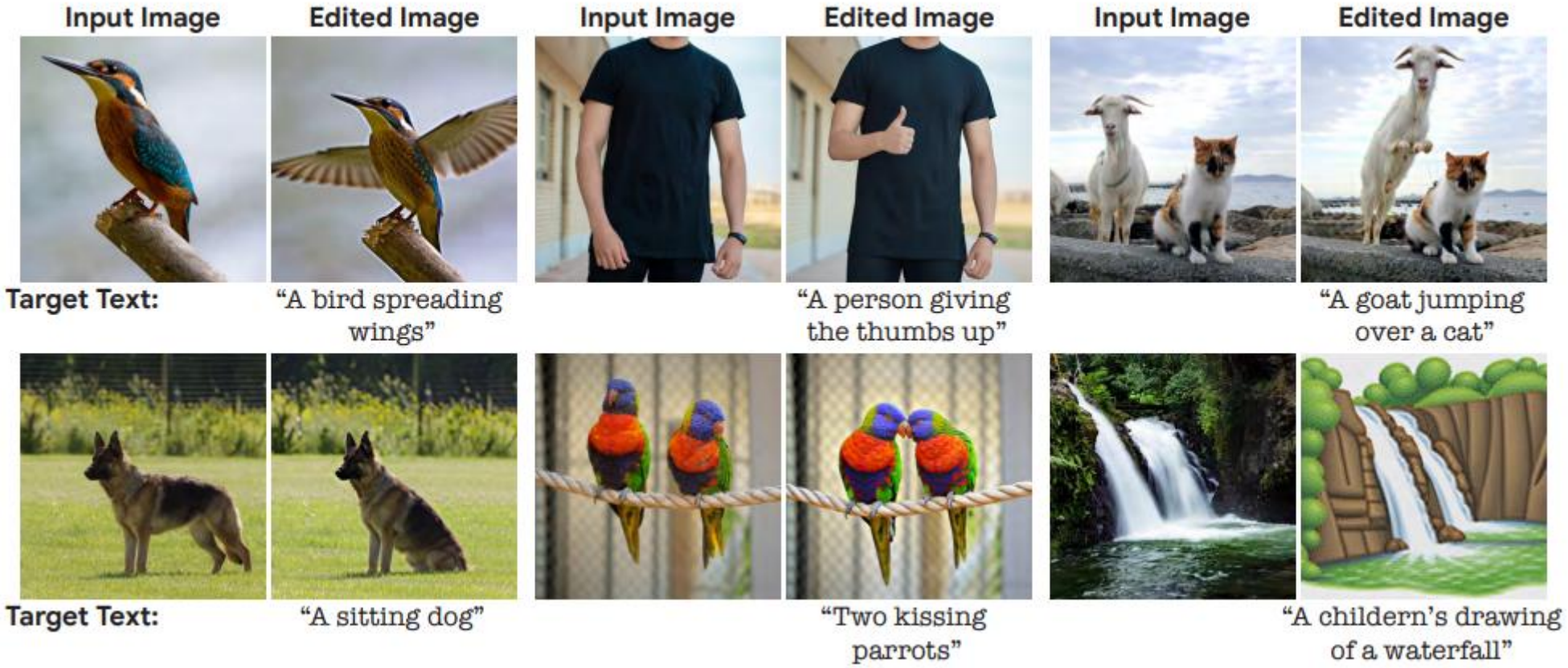
- DiffEdit[16](Meta): Masks utilized.



Selected Papers

Attempts of CDMs on Image Editing: How people flood in a hot direction

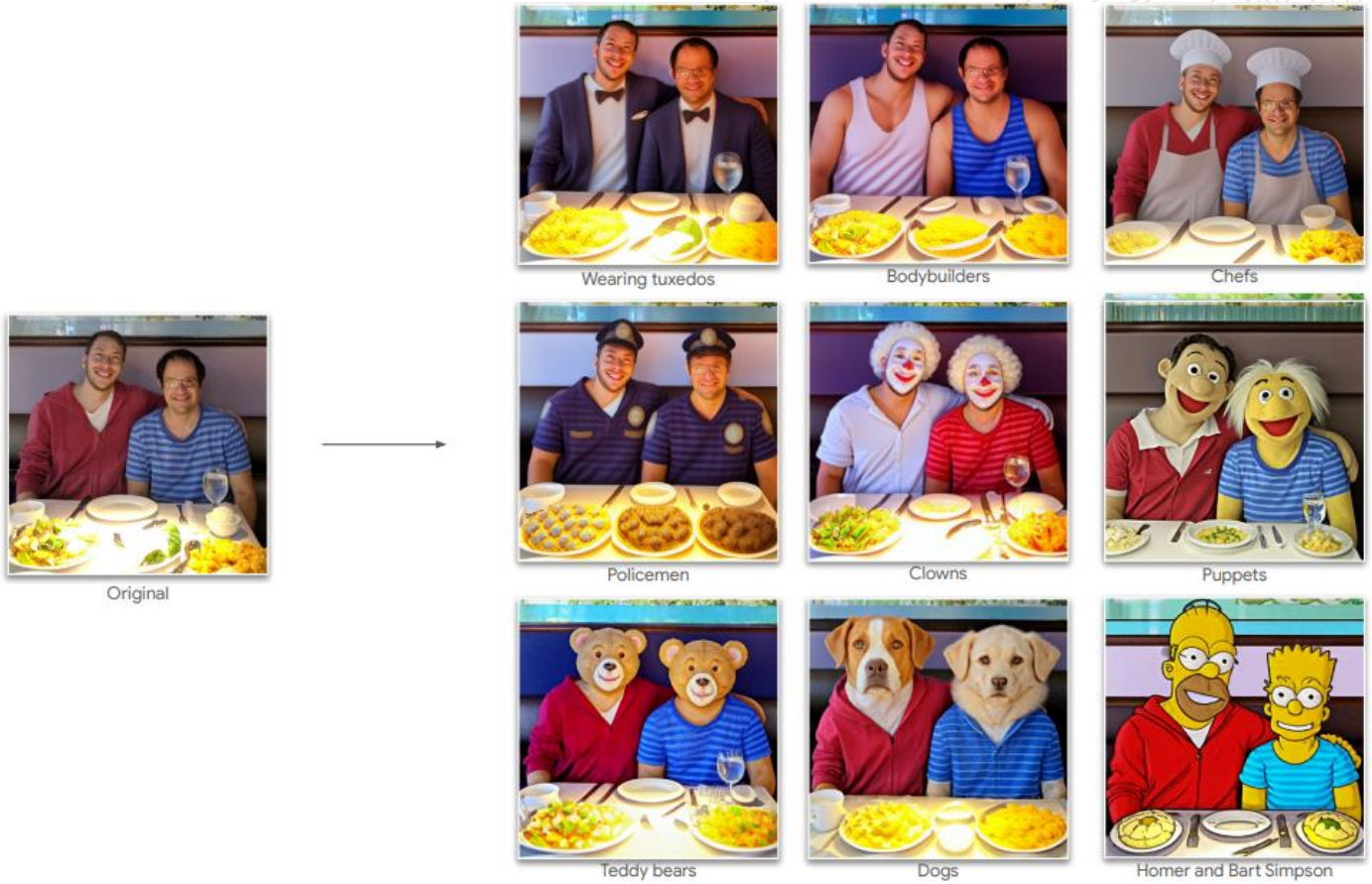
- Imagic:[17](Google): Actions and postures.



Selected Papers

Attempts of CDMs on Image Editing: How people flood in a hot direction

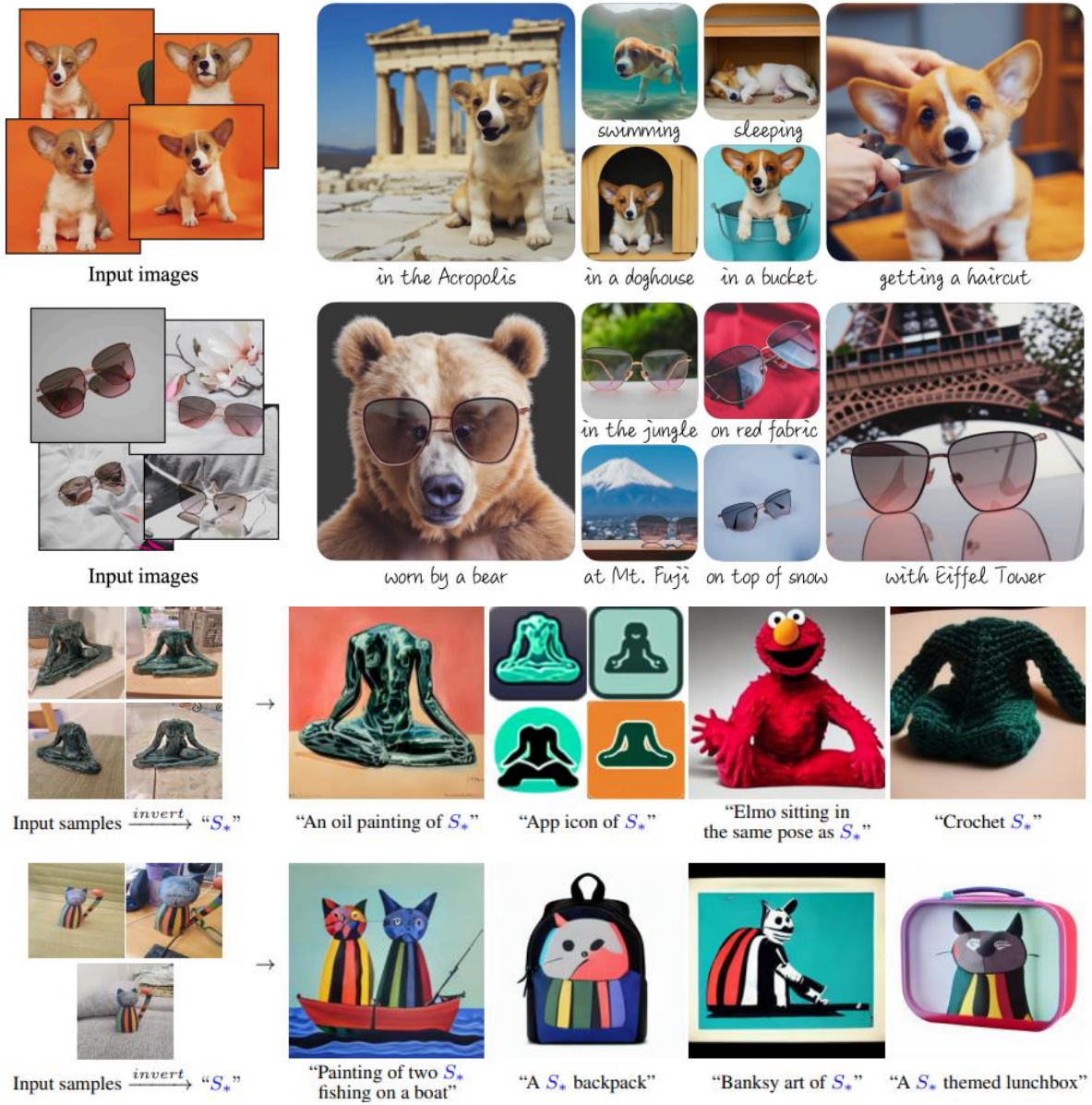
- Unitune:[18](Google): High semantic fidelity.



Selected Papers

Fine-tuning CDMs: Personalized Generation

- DreamBooth[19](Google)



- Textual Inversion[20](Tel Aviv Uni.)

Industrial Application

- OpenAI: DALLE2
- Disco Diffusion; Dance Diffusion(Audio)
- Stability.AI: Stable Diffusion, Joint operation with Novel-AI.

Stability AI, the startup behind Stable Diffusion, raises $101M

Kyle Wiggers @kyle_l_wiggers / 1:01 AM GMT+8 • October 18, 2022

Comment

portrait photo of a african old warrior chief, tribal panther make up, gold on white, side profile, looking away, serious eyes, 50mm portrait photography, hard rim lighting photography–beta –ar 2:3 –beta

The Legend of Zelda landscape atmospheric, hyper realistic, 8k, epic composition, cinematic, octane render, artstation landscape vista photography by Carr Clifton & Galen Rowell, 16K resolution, Landscape veduta photo by Dustin Lefevre & tdraw, 8k resolution, detailed landscape painting by Ivan Shishkin, DeviantArt, Flickr, rendered in Enscape, Miyazaki, Nausicaa Ghibli, Breath of The Wild, 4k detailed post processing, artstation, rendering by octane, unreal engine —ar 16:9

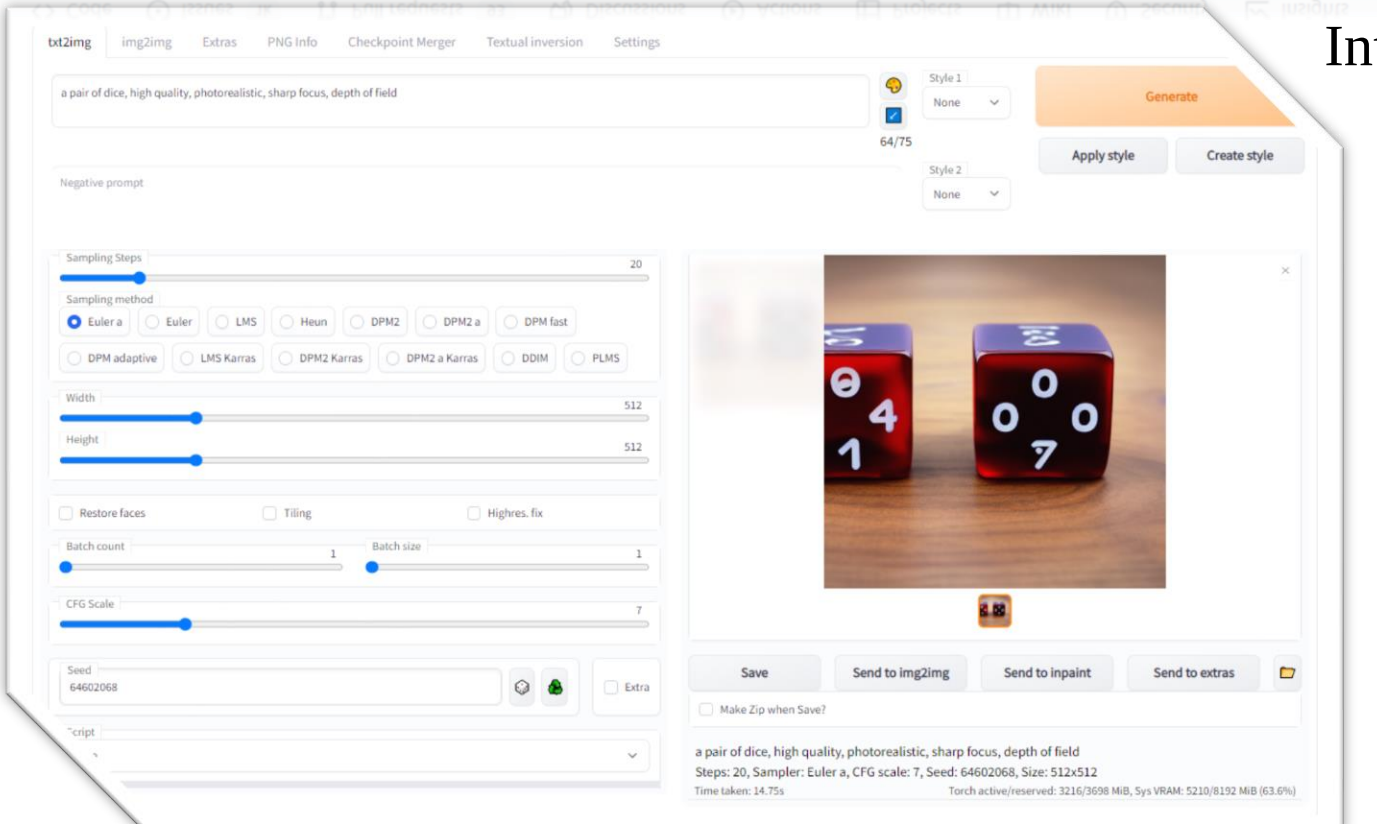
Industrial Application

- The Geek AUTOMATIC1111: stable-diffusion-webui

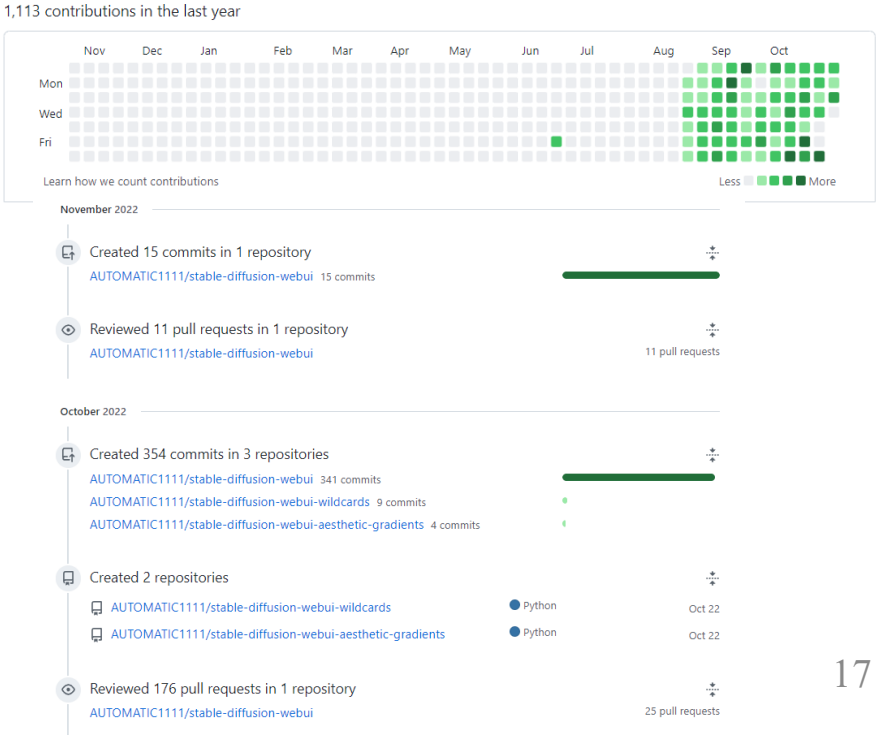
AUTOMATIC1111 / stable-diffusion-webui Public

Watch 195 | Fork 3.3k | Star 18.6k

Code | Issues 1k | Pull requests 93 | Discussions | Actions | Projects | Wiki | Security | Insights

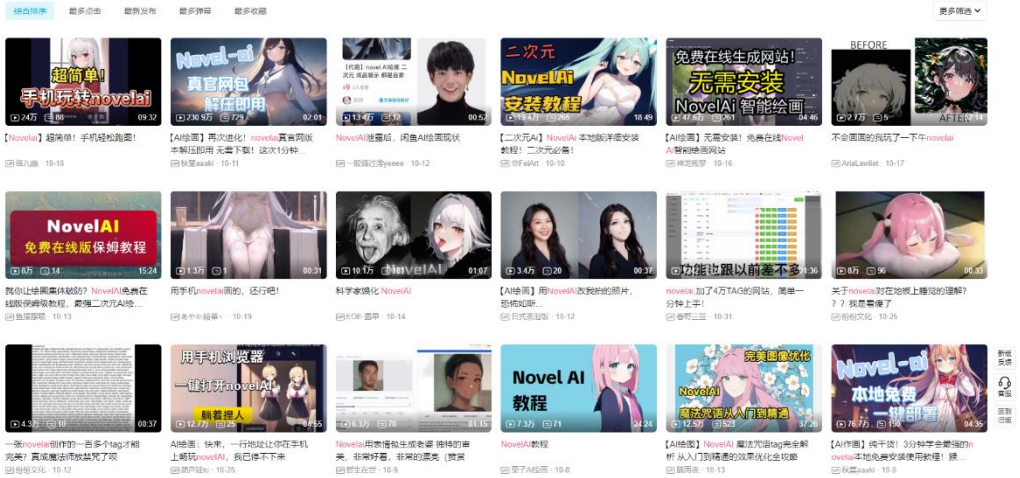


Integrated Deepdanbooru for anime diffusers.

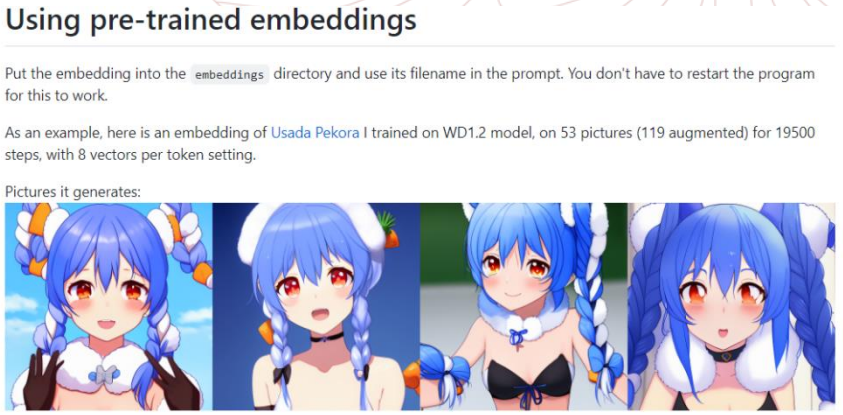


Industrial Application

- The Geek AUTOMATIC1111: stable-diffusion-webui
- [Custom Script]



Img-2-Img



Textual Inversion

- Some interesting community projects
- [Stable diffusion videos]
- [Pokémon Diffusion]

Tools for Research

Huggingface package: [diffusers]

- The latest checkpoint SD V1-5 released on 10.21.
- A lot of easy-to-use code examples provided.

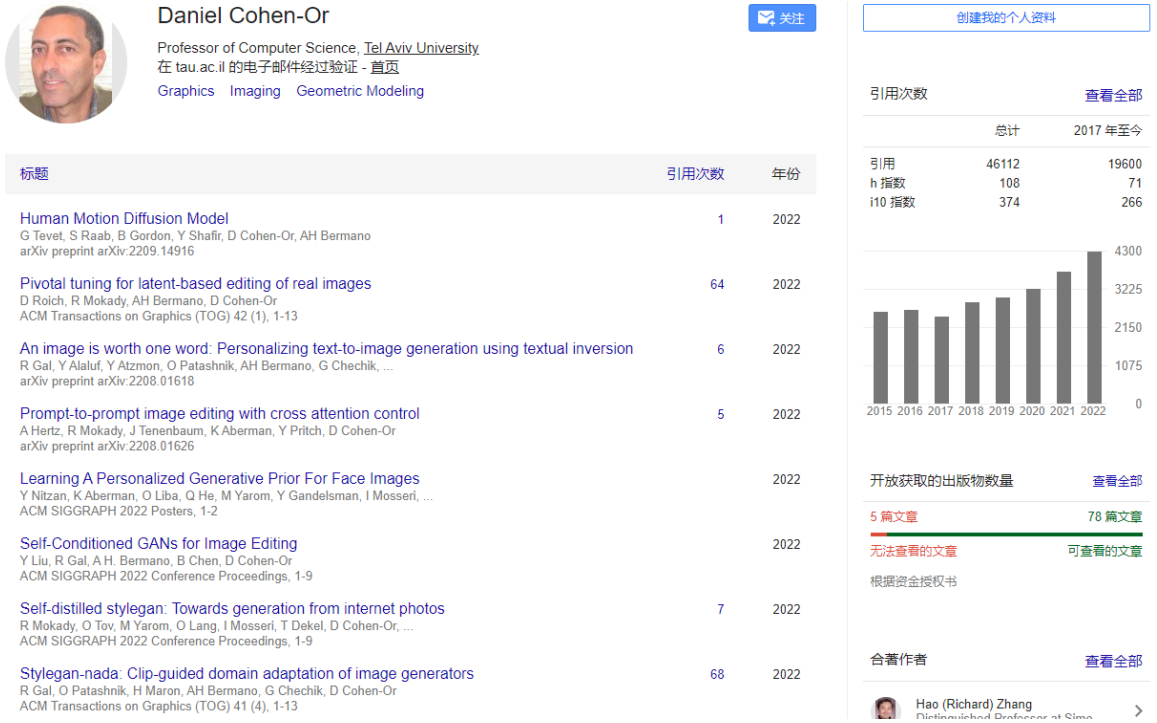
Source Code:
[https://github.com/CompVis/stable-diffusion]

Surveys:

- [Diffusion Models: A Comprehensive Survey of Methods and Applications][21]
- A Survey on Generative Diffusion Model[22]

Group worth following in this direction:

- Daniel Cohen-Or at Tel Aviv University.



Daniel Cohen-Or

关注

Professor of Computer Science, Tel Aviv University
在 tau.ac.il 的电子邮件经过验证 - 首页
Graphics Imaging Geometric Modeling

标题	引用次数	年份
Human Motion Diffusion Model G Tevet, S Raab, B Gordon, Y Shafir, D Cohen-Or, AH Bermano arXiv preprint arXiv:2209.14916	1	2022
Pivotal tuning for latent-based editing of real images D Roich, R Mokady, AH Bermano, D Cohen-Or ACM Transactions on Graphics (TOG) 42 (1), 1-13	64	2022
An image is worth one word: Personalizing text-to-image generation using textual inversion R Gal, Y Alaluf, Y Atzmon, O Patashnik, AH Bermano, G Chechik, ... arXiv preprint arXiv:2208.01618	6	2022
Prompt-to-prompt image editing with cross attention control A Hertz, R Mokady, J Tenenbaum, K Aberman, Y Pritch, D Cohen-Or arXiv preprint arXiv:2208.01626	5	2022
Learning A Personalized Generative Prior For Face Images Y Nitzan, K Aberman, O Liba, Q He, M Yarom, Y Gandelsman, I Mosseri, ... ACM SIGGRAPH 2022 Posters, 1-2		2022
Self-Conditioned GANs for Image Editing Y Liu, R Gal, A H. Bermano, B Chen, D Cohen-Or ACM SIGGRAPH 2022 Conference Proceedings, 1-9		2022
Self-distilled stylegan: Towards generation from internet photos R Mokady, O Tov, M Yarom, O Lang, I Mosseri, T Dekel, D Cohen-Or, ... ACM SIGGRAPH 2022 Conference Proceedings, 1-9	7	2022
Stylegan-nada: Clip-guided domain adaptation of image generators R Gal, O Patashnik, H Maron, AH Bermano, G Chechik, D Cohen-Or ACM Transactions on Graphics (TOG) 41 (4), 1-13	68	2022

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Issues

- Tort
- Privacy and Fairness
- Data Leakage
- NSFW Contents
- Impact on Painters

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Thank you!

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